

Designing Effective Inclusive Supports

Inclusive Schedules

Creating schedules that make supports available, when needed, in inclusive settings is essential to ensuring student progress. This requires an integration of teacher, student, support staff and related services providers schedules. Considering the number of factors to include in the schedule for a single student, the task of designing schedules for a caseload of students becomes quite complex! Generally speaking, then, it is recommended that educators work together to design schedules for students that maximize available resources.

To start, consider the routines that the general education classroom students will participate in, and work with educators to make sure that instructional times when students need support are staggered. In a case study of an effective elementary school, McLeskey et al. (2014) found that teachers provided instruction in core areas (e.g., literacy) at different times of the day, enabling co-teachers and other support staff to be present. Without this simple planning activity, it is quite possible all elementary teachers would teach literacy, for example, at the same time, making it impossible for special educators to be present during literacy instruction in all classrooms.

With this master schedule of general education routines in place, the next step is layering in adult supports. Educators should identify when to co-plan, when to co-teach, and which class activities will have supports of a class-wide paraprofessional. In doing so, adult supports are provided when needed, while other natural supports (e.g., peer-assisted learning) are available for other instructional routines.

Finally, work with related services providers (e.g., speech and language pathologists) to decide on appropriate times and manners for the delivery of these supports. These supports may also include co-teaching; for example, a speech language pathologist may provide small group instruction during a small group activity to teach social communication skills. Other times, the related services provider may need to provide direct instruction in a separate setting due to the nature of the service. However, because pull-out services disrupt learning and account for hours of lost instructional time transitioning to separate areas and missing ongoing instruction, they should be minimized and carefully planned.

Inclusive Instructional Plans

Upon completion of schedules, the next crucial step in delivering effective inclusive instruction is planning what to teach and when to teach it.

What to Teach. What to teach would be based on the general education curriculum and student IEP goals. While this may seem rather straightforward, educators are often challenged to find ways to help students with disabilities "keep up" in both the general education and all IEP goals. Thus, we offer two tips:

- *Ensure IEP goals are linked to grade-appropriate state standards as much as possible.* This will prevent IEP goals from competing with standards for time and instruction. A variety of strategies exist for relating IEP goals to standards, including ensuring that (a) the student's grade level is the point of reference and (b) the goals have centrality with the content and performance expectations of the standard (Browder et al., 2006).
- *Prioritize learning as part of a data-based, team decision-making process.* A critical goal of education is that students keep learning. Often, however, we become overly focused on keeping up. Thus, educators must make decisions that ensure students keep learning, even if they do not keep up with the pace of instruction in the general education classroom. One option is to work with the student's team to prioritize learning content for each unit of instruction (Giangreco et al., 2011). For example, a high school biology class may learn 50 vocabulary words and another 20 key concepts in a unit of study focused on cell meiosis and mitosis. For some students, this is the appropriate number of words and concepts to learn. Other students should learn a different number of concepts. Perhaps a student with significant cognitive disability will learn five vocabulary words and two concepts. The team will decide which of those words and concepts have the most relevance to the student and ensure that curricular adaptations and supports focus on those words and concepts. This allows the student to keep learning grade-aligned academic content, without the pressure of keeping up. As students receive

instruction and educators collect data, teams can adjust the curricular goals for a student during a lesson based on rate of learning, as needed.

When to Teach. With grade-aligned IEP goals developed and instructional goals identified for a unit of study, educators can then address the next concern: when to teach. Far too often, when to teach has been conflated with where to teach. That is, often students have goals that differ substantially from what is taught in general education, and teams have no clear instructional goals for learning general education content. As a result, students are pulled out of general education classrooms and taught skills in special education settings.

This is avoidable not only by using the strategies discussed previously but also by using embedded instruction, in an evidenced-based practice (Jimenez & Kamei, 2015) that involves overlaying general education routines with student IEP goals and instructional priorities. Teams determine when, in a typical day, students will have natural opportunities to practice IEP goals and instructional priorities. They then determine if students have enough opportunities to practice skills in a typical day. If not enough opportunities are present (based on student progress monitoring data), additional supplemental opportunities are provided. For example, math is a natural time to work on addition and subtraction IEP goals, and small-group work is a natural time to work on social communication goals.

The instructional priorities identified for general education classes will have many natural opportunities to practice them during most class days. However, some students have IEP goals that are necessary for a student but may not occur often enough during a typical school day. For example, perhaps a high school student with significant cognitive disability is learning to identify coins by name and value. Because most high school students have mastered this skill, it is not taught in high school courses. The only natural times to practice coins would be in the school snack bar, but this still presents too few opportunities. The team could then embed supplemental teaching opportunities for coin identification during noninstructional moments throughout the day, including while the teacher is taking attendance, when students are getting out materials, or transitioning between activities. During these times, a peer partner, paraprofessional, or teacher could present an instructional opportunity (e.g., "which one is five cents?" while hold-

ing two coins). This could be repeated a few times throughout the day, enabling students to practice a variety of skills (McDonnell et al., 2008) without having to leave the general education classroom or miss instruction to practice learning priority skills.

Technology in Inclusive Classrooms

Personal technology today is ubiquitous: smartphones, tablets, and voice assistants in our homes that can do everything from answer questions to turn on lights—in short, technology has a profound impact on daily life today. This same technology is also transforming how we provide instruction. Students have access to supports such as speech-to-text software, screen-reading programs, voice-output communication devices, and many others in ways unheard of just a few years ago. As these technologies gain users and simplify life for people around the world, their cost continues to decrease, and their availability in schools spreads. It is essential that students with and without disabilities have access to and opportunities to learn with these technologies, as appropriate, not only to prepare them to be citizens of the world today but also to facilitate learning.

Universal Design for Learning

Like any personalized support, students are not born knowing what technology suits them best, or how to use technology. Instruction and opportunity must therefore be provided to students on a regular basis, enabling them to learn what works best for them and choose to use specific technologies for specific purposes. Inclusive educators must embed technology in their classrooms, using strategies such as uni-

versal design for learning (UDL; Cook & Rao, 2018).

As discussed in Chapter 2,

UDL is based on the prem-

ise that all instruction should be designed from the beginning to reduce barriers, and technology provides ready means of reducing barriers. For example, in a literacy class in which students are learning text comprehension through reading a novel, educators might provide several forms of the novel: in print

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form, in audio form via text-to-speech software, and simplified text enhanced with images.

Using a UDL approach, students choose which format they prefer to learn from and are held to the same text comprehension expectations regardless of form.

Technology-Assisted Learning

Technology is also useful in making learning fun and engaging. Often educators fear technology will be distracting to students, yet many students prefer to use technology to learn because they believe it makes learning more fun and interesting. Difficult or "boring" subjects can come to life through virtual lessons, videos, and interactive games. Importantly, technology-assisted learning (learning on tablets or computers) can also facilitate personalized learning. The use of these technologies reduces stigma and enables students to work on materials that are appropriate to their skill level using the supports they need (e.g., large font or text-to-speech). Technology can similarly enable students to learn at their own pace through this personalized instruction. Research has documented how various apps can be used to teach diverse skills, such as phonics to students who use speech-generating applications on iPads (Ahlgrim-Delzell et al., 2016) and writing for students with cognitive disability (Felix et al., 2017). In sum, integration of technology in the inclusive classroom has the capacity of technology to open doors for student learning, communicating, and personalized supports in ways that were impossible in previous generations.

Conclusion

In this chapter we have taken a necessarily broad look at answering the question, How can we teach better? These effective practices emerged from second-generation inclusive practices (discussed in greater detail in Chapter 5) in which students with disabilities were increasingly being included in the general education classroom, and teachers needed to know how to teach them in that context. Each practice could be a book in and of itself, and future volumes of the Norton Inclusive Education for Students With Disabilities series will expand on some of these. It is important to understand that evidence-based and high-leverage practices have been shown to be important to inclusive education.

Reflective Questions for Group Study

1. How do family-school partnerships, instructional strategies, teaching arrangements, teaching strategies, and technology each raise expectations and promote positive student outcomes?
2. In what ways do educator roles and responsibilities change in an inclusive classroom?
3. How do the instructional cycles described in this chapter create an opportunity for continuous evaluation and improvement of inclusive instruction?
4. Describe how application of the Universal Design for Learning framework would improve inclusive opportunities for all students.